Sahaj Bhadja

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Summary

Software engineer with 2+ years of experience deploying **AI-powered systems** across **Python**, **TypeScript**, **SQL**, and cloud platforms. Strong in client-facing implementation, API integration, prompt-oriented LLM workflows, troubleshooting, and shipping secure, explainable, production-grade solutions from requirements through go-live.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University **Gandhinagar, In**
Bachelor of Technology in Computer Science and Engineering *November 2020 – May 2*
GPA: 3.88/4 Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Architected a multi-tenant **AI platform** across **Azure SQL**, **PostgreSQL**, and cloud storage for 500+ users, reducing dashboard latency from **1.2s to 350ms** through schema, indexing, and query optimization.
- Built **FastAPI** integration services and fault-isolated LLM orchestration workflows for **5K+ monthly jobs**, adding validation, release checks, and production observability to sustain **99%+ reliability**.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Engineered an AI-powered **Flask + React** grading platform using multimodal LLMs and OCR to evaluate **10K+ submissions**, translating instructor workflows into production features and prompt-driven grading pipelines.
- Implemented concurrent processing, secure file-ingestion, and **AWS RDS**-backed REST APIs with **GitHub Actions CI/CD**, handling **10K+ grading requests/day** while cutting turnaround time by **60%**.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Served as a technical implementation partner for Microsoft/GitHub stakeholders, configuring cross-platform microservice software and automated test flows with **Jenkins**, PyTest, and Mockito across **3 operating systems**.
- Troubleshot API and ETL issues using **OOP patterns**, **AWS Glue**, and S3, improving response times by **30%**, pipeline efficiency by **68%**, and reducing technical debt by **20%**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Worked directly with the client to map **12+ backend integration points**, define end-to-end booking and payment workflows, and translate requirements into Flask APIs and real-time Firebase-backed synchronization across **50K+ records**.
- Integrated webhook-driven payment flows and secure authentication patterns, reducing manual payment handling by **70%**, improving confirmation speed by **40%**, and strengthening production readiness.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a **MERN** platform with **REST APIs**, **WebSockets**, and microservices, supporting **8+ business workflows** and **2M+ synchronized records** in a user-facing operations system.
- Automated deployments and validation with **Jenkins**, Docker, and Azure SQL-backed services, reducing deployment time by **50%** while reaching **95%+ successful build validation**.

Skills

Languages: Python, JavaScript/TypeScript, Java, SQL, C++
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, JSON, Prompt-Oriented LLM Workflows, System Configuration
Frontend: React.js, Next.js, TailwindCSS, HTML, CSS
Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Azure SQL, Firebase
Cloud & Infrastructure: GCP, AWS (RDS, S3, Glue), Microsoft Azure, Docker, Kubernetes, Jenkins, GitHub, Linux
AI/ML: LLM Systems, GPT-4o, Claude, OCR Workflows, Prompt Engineering, Applied AI Integrations
Engineering Practices: CI/CD, Troubleshooting, Documentation, Microservices, ETL Pipelines, Agile, System Design, Secure Scalable Systems

Projects

Big-Brain | *LLaMA-2, FAISS, PostgreSQL, GCP, REST APIs*
- Built an AI document platform over **50K+ documents** with semantic search, summarization, secure upload workflows, and cloud-based indexing pipelines, reducing retrieval latency by **40%** for **200+ users**.

Chat2SQL | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting **10+ databases**, accelerating query authoring, schema exploration, and structured execution across a **Java + TypeScript** product surface.

CaptionCraft | *Transformers, PyTorch, Flask, LLMs*
- Benchmarked and deployed transformer-based captioning APIs across a **10K image / 100K caption** dataset, improving caption accuracy by **15%** and strengthening prompt- and model-driven inference workflows.